Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

SQL Masterclass

Course Code: C024

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 October 2026	Initial release of the SQL Masterclass lesson plan: 1 day, 7.5 instructional hours (9:30 am – 5:30 pm), 11 hands-on labs across 4 topics.	Dr. Alfred Ang

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TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Course Schedule **4**
Day 1 — Data Modeling, SQL Queries, Transformation & Data Warehouse 4
Lab Reference **6**

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Course Information

Course Title	SQL Masterclass
Course Code	C024
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 7.5 instructional hours
Daily Timing	9:30 am – 5:30 pm (30-minute lunch break)
Mode	Instructor-led, hands-on SQL labs in SQLite Studio on the SG Mart sample database (sgmart.db)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply data modeling for business processes.
- LO2: Apply data processing and analysis using SQL.
- LO3: Apply data transformation from multiple data sources.
- LO4: Apply data mapping to data warehouse.

Course Schedule

Day 1 — Data Modeling, SQL Queries, Transformation & Data Warehouse

Time	Duration	Topic / Activity
9:30–10:10	40 min	Welcome, course introduction, trainer and learner introductions, ground rules, the SG Mart course dataset, lesson plan and course outline. Slides 1–17
10:10–11:30	80 min	**SQL Foundations: what SQL is and can do, history, SQL databases, tables, SQLite Studio. Topic 1 — Data Modeling: CREATE DATABASE/TABLE, data types, constraints, keys and data modeling (concepts + demo). Hands-on: Lab 1: Set Up SQLite Studio & Import the SG Mart Database; Lab 2: Create a Database and Tables. Slides 18–45**
11:30–13:00	90 min	Hands-on: Lab 3: Model Data with Constraints and Keys. Topic 2 — Data Processing and Analysis: SELECT, DISTINCT, WHERE, operators, ORDER BY, LIMIT, aliases (concepts + demo). Hands-on:

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		Lab 4: Query Data with SELECT. Slides 46–64
13:00–13:30	30 min	Lunch break
13:30–14:50	80 min	INSERT, UPDATE, DELETE (concepts + demo). Hands-on: Lab 5: Filter with SQL Operators; Lab 6: Insert, Update and Delete Records. Slides 66–72
14:50–16:10	80 min	Topic 3 — Data Transformation: aggregate functions, joins, UNION, GROUP BY, HAVING (concepts + demo). Hands-on: Lab 7: Aggregate Data with COUNT, AVG and SUM; Lab 8: Join Data from Multiple Tables; Lab 9: Group Data with GROUP BY and HAVING. Slides 73–93
16:10–17:10	60 min	Topic 4 — Introduction to Data Warehouse: warehouse concepts, OLAP, RDBMS, E-R model, SMRT case study, stored procedures (concepts + demo). Hands-on: Lab 10: Map an E-R Model to Database Tables (SMRT Case Study); Lab 11: Author Stored Procedures for SQL Server. Slides 94–113
17:10–17:30	20 min	Course recap and summary, Q&A, recommended courses and course feedback. Slides 114–119

Total instructional time: 450 minutes (7.5 hours).

Lab Reference

Topic	Labs
Topic 01: Data Modeling	Lab 1: Set Up SQLite Studio & Import the SG Mart Database; Lab 2: Create a Database and Tables; Lab 3: Model Data with Constraints and Keys
Topic 02: Data Processing and Analysis	Lab 4: Query Data with SELECT; Lab 5: Filter with SQL Operators; Lab 6: Insert, Update and Delete Records
Topic 03: Data Transformation	Lab 7: Aggregate Data with COUNT, AVG and SUM; Lab 8: Join Data from Multiple Tables; Lab 9: Group Data with GROUP BY and HAVING
Topic 04: Introduction to Data Warehouse	Lab 10: Map an E-R Model to Database Tables (SMRT Case Study); Lab 11: Author Stored Procedures for SQL Server

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